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Q.1 If $u = f(r)$, where $r^2 = x^2+y^2$, then show that
$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = f''(r) + \frac{1}{r} f'(r)$$

Sol^n →
$$u = f(r) \quad \text{--- (1)}$$
$$r^2 = x^2+y^2 \quad \text{--- (2)} \rightarrow d.w.r. \text{ to } x \Rightarrow dr/dx = x/r$$

Eq^n (1) d. w. r. to 'x' (P)
$$\frac{\delta u}{\delta x} = f'(r) \times \frac{\delta r}{\delta x} \Rightarrow \frac{\delta u}{\delta x} = \frac{x}{r} f'(r) \quad \text{--- (3)}$$

Diff. (3) w. r. to 'x' (P)
$$\frac{\delta^2 u}{\delta x^2} = \frac{r \times \frac{\delta}{\delta x}[x f'(x)] - x f'(r) \times \frac{\delta r}{\delta x}}{r^2}$$
$$\frac{\delta^2 u}{\delta x^2} = \frac{r\{1 \times f'(r) + x\, f''(r) \times (\delta r/\delta x)\} - x\, f'(r) \times x/r}{r^2}$$
$$\frac{\delta^2 u}{\delta x^2} = \frac{r\{f'(r) + x^2/r\, f''(r)\} - x^2/r\, f'(r)}{r^2}$$
$$\frac{\delta^2 u}{\delta x^2} = \frac{f'(r)}{r} + \frac{x^2}{r^2} f''(r) - \frac{x^2}{r^3} f'(r) \quad \text{--- (A)}$$

Similarly,
$$\frac{\delta^2 u}{\delta y^2} = \frac{f'(r)}{r} + \frac{y^2}{r^2} f''(r) - \frac{y^2}{r^3} f'(r) \quad \text{--- (B)}$$

Adding Equation (A) and (B)

$$\boxed{\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = \frac{f'(r)}{r} + f''(r)}$$

Hence Proved.

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Q.2 If $u = \log\left(\frac{x^3+y^3}{x^2+y^2}\right)$, then prove that $x\frac{\delta u}{\delta x} + y\frac{\delta u}{\delta y} = 1$

Solⁿ → $u = \log\left(\frac{x^3+y^3}{x^2+y^2}\right)$

Taking 'e' on both side

$e^u = \frac{x^3+y^3}{x^2+y^2}$

$e^u = \frac{x^3(1+(y/x)^3)}{x^2(1+(y/x)^2)}$

$\boxed{e^u = x\phi\left(\frac{y}{x}\right)}$

e^u is the homogenous function of 1 degree in x and by Euler's theorem —

$x\frac{\delta}{\delta x}(e^u) + y\frac{\delta}{\delta y}(e^u) = 1e^u$

$xe^u\frac{\delta u}{\delta x} + ye^u\frac{\delta u}{\delta y} = e^u$

$e^u\left[x\frac{\delta u}{\delta x} + y\frac{\delta u}{\delta y}\right] = e^u$

$x\frac{\delta u}{\delta x} + y\frac{\delta u}{\delta y} = \frac{e^u}{e^u}$

$\boxed{x\frac{\delta u}{\delta x} + y\frac{\delta u}{\delta y} = 1}$ H.P.

If $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, Prove that $x\frac{\delta u}{\delta x} + y\frac{\delta u}{\delta y} = \tan u$

$u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$

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Taking 'sin' on both side

$\sin u = \frac{x^2(1+(y/x)^2)}{x(1+(y/x))}$

$\sin u = x\phi\left(\frac{y}{x}\right)$

$\sin u$ is the homogenous function of 1 degree in x and by Euler's theorem —

$x\frac{\delta}{\delta x}(\sin u) + y\frac{\delta}{\delta y}(\sin u) = 1\sin u$

$x\cos u\frac{\delta u}{\delta x} + y\cos u\frac{\delta u}{\delta y} = \sin u$

$\cos u\left[x\frac{\delta u}{\delta x} + y\frac{\delta u}{\delta y}\right] = \sin u$

$x\frac{\delta u}{\delta x} + y\frac{\delta u}{\delta y} = \frac{\sin u}{\cos u}$

$\boxed{x\frac{\delta u}{\delta x} + y\frac{\delta u}{\delta y} = \tan u}$ H.P.

Q.4 If $x^x y^y z^z = c$, show that $x=y=z$, $\frac{\delta^2 z}{\delta x\delta y} = -[x\log ex]^{-1}$

Solⁿ → $x^x y^y z^z = c$ — (1)

Taking 'log' on both side

$\log(x^x y^y z^z) = \log c$

$x\log x + y\log y + z\log z = \log c$ — (2)

d.w.r. x. to 'x' (p) and z is the function of x, y

$x\cdot\frac{1}{x} + \log x + 0 + z\cdot\frac{1}{z} + \log z\frac{dz}{dx} = \log c$

$(1+\log x) + (1+\log z)\frac{dz}{dx} = \cancel{\log c}\ 0$

$\frac{\delta z}{\delta x} = -\frac{(1+\log x)}{(1+\log z)}$ — (3)

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$x\log x + y\log y + z\log z$

d. w. r. to 'y' (Eq^n 2)

$$0 + y\cdot\frac{1}{y} + \log y + z\cdot\frac{1}{z} + \log z\frac{\delta z}{\delta y} = 0$$

$$(1+\log y) + (1+\log z)\frac{\delta z}{\delta y} = 0$$

$$\frac{\delta z}{\delta y} = -\frac{(1+\log y)}{(1+\log z)} \quad \text{---} (4)$$

Diff. w. r. to y (Partially)

$$\frac{\delta^2 z}{\delta x\delta y} = -(1+\log x)\times\frac{\delta}{\delta y}\left[\frac{1}{(1+\log z)}\right]$$

$$= -(1+\log x)\times\frac{\delta}{\delta y}\left[(1+\log z)^{-1}\right]$$

$$= -(1+\log x)\times\left[-(1+\log z)^{-2}\times\left(0+\frac{1}{z}\times\frac{\delta z}{\delta y}\right)\right]$$

$$\frac{\delta^2 z}{\delta x\delta y} = \frac{(1+\log x)}{(1+\log z)^2}\times\frac{1}{z}\times\frac{\delta z}{\delta y}$$

$$\frac{\delta^2 z}{\delta x\delta y} = \frac{-(1+\log x)}{z(1+\log z)^2}\times\left[\frac{(1+\log y)}{(1+\log z)}\right]$$

$$\frac{\delta^2 z}{\delta x\delta y} = \frac{-(1+\log x)(1+\log x)}{x(1+\log x)^3}$$

$$= \frac{-1}{x(1+\log x)} = -\left[x(1+\log x)\right]^{-1}$$

$$\boxed{\frac{\delta^2 z}{\delta x\delta y} = -\left[x(1+\log x)\right]^{-1}}$$

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Que. 5 Solve the following differential equations :

$$\frac{d^2y}{dx^2} + 4y = x\cos x$$

A.E $\Rightarrow$ $m^2 + 4 = 0$

$m^2 = -4$

$m = \pm 2i$

$\boxed{m = 0 \pm 2i}$

C.F = $c_1\cos 2x + c_2\sin 2x$

$$PI = \frac{1}{D^2+4}(x\cos x)$$

$$= x\times\frac{1}{D^2+4}\cos x - \frac{2D}{(D^2+4)^2}\cos x$$

$$= x\times\frac{1}{-1+4}\cos x - \frac{2D}{(-1+4)^2}\cos x$$

$$= \frac{1}{3}x\cos x - \frac{2D}{9}\cos x$$

$$PI = \frac{1}{3}x\cos x + \frac{2}{9}\sin x$$

$y = CF + PI$

$$\boxed{y = c_1\cos 2x + c_2\sin x + \frac{1}{3}x\cos x + \frac{2}{9}\sin x}$$

Que. 6 Solve the following differential equation $(D^2+3D+2)y = x\sin 2x$

$\Rightarrow$ $(D^2+3D+2)y = x\sin 2x$

A.E. $\Rightarrow$ $m^2 + 3m + 2 = 0$

$m^2 + 2m + m + 2$

$m(m+2) + 1(m+2)$

$(m+1)(m+2) = 0 \Rightarrow \boxed{m = -1, -2}$

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$$C.F. = c_1 e^{-x} + c_2 e^{-2x}$$

$$PI = \frac{1}{D^2+3D+2} x\sin 2x$$

$$= x \times \frac{1}{D^2+3D+2}\sin 2x - \frac{2D+3}{(D^2+3D+2)^2}\sin 2x$$

$$= x \times \frac{1}{-4+3D+2}\sin 2x - \frac{2D+3}{(-4+3D+2)^2}\sin 2x$$

$$= x \times \frac{1}{-2+3D}\sin 2x - \frac{2D+3}{(-2+3D)^2}\sin 2x$$

$$= x \times \frac{(3D+2)}{(3D-2)(3D+2)}\sin 2x - \frac{2D+3}{9D^2+4-12D}\sin 2x$$

$$= \frac{x(3D+2)}{9D^2-4}\sin 2x - \frac{2D+3}{-36+4-12D}\sin 2x$$

$$= \frac{x(3D+2)}{-36-4}\sin 2x - \frac{2D+3}{-32-12D}\sin 2x$$

$$= \frac{-1}{40}x(3D+2)\sin 2x - \frac{2D+3}{-4(3D+8)}\sin 2x$$

$$= \frac{-1}{40}x(2\sin 2x+6\cos 2x) + \frac{1}{4}\frac{2D+3}{(3D+8)}\frac{(3D-8)}{(3D+8)}\sin 2x$$

$$= \frac{-1}{40}x(2\sin 2x+6\cos 2x) + \frac{1}{4}\frac{6D^2-16D+9D-24}{(9D^2-64)}\sin 2x$$

$$= \frac{-1}{40}x(2\sin 2x+6\cos 2x) + \frac{1}{4}\frac{(6D^2-7D-24)}{9(-4)-64}\sin 2x$$

$$= \frac{-1}{40}x(2\sin 2x+6\cos 2x) + \frac{1}{4}\frac{6(-4)-7D-24}{(-100)}\sin 2x$$

$$= \frac{-1}{40}x(2\sin 2x+6\cos 2x) \frac{-48-7D}{-400}\sin 2x$$

$$= \frac{-1}{40}x(2\sin 2x+6\cos 2x) + \frac{1}{400}(48+7D)\sin 2x$$

$$PI = \frac{-1}{40}x(2\sin 2x+6\cos 2x) + \frac{1}{400}(48\sin 2x + 14\cos 2x)$$

$$y = CF + PI$$

$$\boxed{y = c_1 e^{-x} + c_2 e^{-2x} + \left(\frac{-1}{40}\right)x(2\sin 2x+6\cos 2x) + \frac{1}{400}(48\sin 2x+14\cos 2x)}$$

Q.7 Solve the following differential equation

$$(1+y^2)\,dx = (\tan^{-1}y - x)\,dy$$

$$(1+y^2)\,dx = (\tan^{-1}y - x)\,dy$$

$$(1+y^2)\frac{dx}{dy} = \tan^{-1}y - x$$

$$\frac{dx}{dy} = \frac{\tan^{-1}y - x}{1+y^2}$$

$$\frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{\tan^{-1}y}{1+y^2}$$

$$P = \frac{1}{1+y^2}, \quad Q = \frac{\tan^{-1}y}{1+y^2}$$

$$I.F. = e^{\int P\cdot dy}$$

$$I.F = e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$$

$$\boxed{IF = e^{\tan^{-1}y}}$$

General solution ⇒

$$x \times e^{\tan^{-1}y} = \int \frac{\tan^{-1}y}{1+y^2}\cdot e^{\tan^{-1}y}\,dy + c$$

Let $t = \tan^{-1}y$

$$dt = \frac{1}{1+y^2}dy$$

$$= \int t\cdot e^t\,dt = te^t - e^t = e^t(t-1) + c$$

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$$\boxed{x \times e^{\tan^{-1}y} = e^{\tan^{-1}y}(\tan^{-1}y - 1) + c}$$

Q.8 Solve the following differential equation

$$(1-x^2)\frac{dy}{dx} + xy = xy^2$$

$\Rightarrow$ $$(1-x^2)\frac{dy}{dx} + xy = xy^2$$

$$\frac{(1-x^2)}{y^2}\frac{dy}{dx} + \frac{xy}{y^2} = x$$

$$\frac{(1-x^2)}{y^2}\frac{dy}{dx} + \frac{x}{y} = x$$

$$\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{(1-x^2)} \times \frac{x}{y} = x \times \frac{1}{(1-x^2)}$$

$$y^{-2}\frac{dy}{dx} + y^{-1}\frac{x}{(1-x^2)} = \frac{x}{(1-x^2)}$$

Let $V = y^{-1}$

d.w.r. to 'x'

$$\frac{dv}{dx} = -y^{-2}\frac{dy}{dx}$$

$$-\frac{dv}{dx} + \frac{x}{(1-x^2)}V = \frac{x}{(1-x^2)}$$

$$\frac{dv}{dx} - \frac{x}{(1-x^2)}V = \frac{-x}{(1-x^2)}$$

$$P = \frac{-x}{(1-x^2)}, \quad Q = \frac{-x}{(1-x^2)}$$

$$I.F. = e^{\int P \cdot dx}$$

$$I.F. = e^{\int \frac{-x}{(1-x^2)}dx} = e^{\frac{1}{2}\log|1-x^2|} = \sqrt{1-x^2}$$

$$\boxed{I.F. = \sqrt{1-x^2}}$$

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Solution of equation –

$$v \times \sqrt{1-x^2} = \int \frac{-x}{(1-x^2)} \times \sqrt{1-x^2}\, dx$$

$$v \times \sqrt{1-x^2} = \int \frac{-x}{\sqrt{1-x^2}}\, dx$$

let $1-x^2 = t$

$-2x\,dx = dt$

$$v \times \sqrt{1-x^2} = \frac{1}{2}\int \frac{dt}{\sqrt{t}} = \frac{1}{2} \times 2\sqrt{t} + c$$

$$v \times \sqrt{1-x^2} = \sqrt{1-x^2} + c$$

$$\boxed{y^{-1} \times \sqrt{1-x^2} = \sqrt{1-x^2} + c}$$

Q.9 Solve the following differential equation

$$\cos x \frac{d^2y}{dx^2} + \sin x \frac{dy}{dx} - 2y\cos^3 x = 2\cos^5 x$$

$\Rightarrow$ The given differential equation is –

$$\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} - 2y\cos^2 x = 2\cos^4 x \quad \text{—(1)}$$

$$\frac{d^2y}{d\cancel{x}z^2} + P_1\frac{dy}{dz} + Q_1 y = R_1 \quad \text{—(2)}$$

$$P_1 = \frac{\frac{d^2z}{dx^2} + P\frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2}, \quad Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2}, \quad R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2}$$

Let $Q_1 = 2 \Rightarrow -2 = \frac{-2\cos^2 x}{\left(\frac{dz}{dx}\right)^2} \Rightarrow \left(\frac{dz}{dx}\right)^2 = \cos^2 x$

$$\frac{dz}{dx} = \cos x \qquad \text{By } z = \sin x$$

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$P_1 = \frac{-\sin x + \tan x \cos x}{\cos^2 x} = 0$

$R_1 = \frac{2\cos^4 x}{\cos^2 x} = 2\cos^2 x = 2(1-z^2)$

Put in the value of equation (2)

$\frac{d^2y}{dz^2} - 2y = 2(1-z^2)$

~~C.F. = C~~

A.E. $\Rightarrow m^2 - 2 = 0$

$\boxed{m = \pm 2}$

C.F. $= C_1 e^{\sqrt{2}z} + C_2 e^{-\sqrt{2}z}$

$= C_1 e^{\sqrt{2}\sin x} + C_2 e^{-\sqrt{2}\sin x}$

P.I. $= \frac{1}{D^2-2} \cdot 2(1-z^2)$

$= -\left(1 - \frac{D^2}{2}\right)^{-1}(1-z^2)$

$= -\left(1+\frac{D^2}{2}\right)(1-z^2) = -(\cancel{1}-z^2-\cancel{1}) = z^2$

$\boxed{PI = \sin^2 x}$

$\therefore$ $y = CF + PI$

$\boxed{y = C_1 e^{\sqrt{2}\sin x} + C_2 e^{-\sqrt{2}\sin x} + \sin^2 x}$

Q.10 Solve the following differential equation

$x\frac{d^2y}{dx^2} + (4x^2-1)\frac{dy}{dx} + 4x^3 y = 2x^3$

$x\frac{d^2y}{dx^2} + (4x^2-1)\frac{dy}{dx} + 4x^3 y = 2x^3$

$\frac{d^2y}{dx^2} + \left(4x - \frac{1}{x}\right)\frac{dy}{dx} + 4x^2 y = 2x^2$ —①

$\frac{d^2y}{dz^2} + P_1\frac{dy}{dz} + Q_1 y = R_1$ —②

$Q_1 \Rightarrow 4 = \frac{4x^2}{\left(\frac{dz}{dx}\right)^2} \Rightarrow x^2 = \left(\frac{dz}{dx}\right)^2$

$\frac{dz}{dx} = x \Rightarrow z = \frac{x^2}{2}$

$P_1 = \frac{1 + \left(4x - \frac{1}{x}\right) \times x}{x^2} = \frac{1 + \left(\frac{4x^2-1}{x}\right) \times x}{x^2} = 4$

$R_1 = \frac{2x^2}{x^2} = 2$

Put in the value of equation (2)

$\frac{d^2y}{dz^2} + 4\frac{dy}{dz} + 4y = 2x$

A.E. $\Rightarrow m^2 + 4m + 4 = 0$

$m^2 + 2m + 2m + 4$

$m(m+2) + 2(m+2)$

$(m+2)(m+2)$

$\boxed{m = -2, -2}$

C.F. $= (C_1 + C_2 z)e^{-2z} \Rightarrow \left(C_1 + C_2\frac{x^2}{2}\right)e^{-x^2}$

PI $= \frac{1}{D^2+4D+4} R \cdot \Rightarrow \frac{1}{(D+2)^2}(2e^{0z})$

$$P.I. = \frac{1}{(0+2)^2}(2) = \frac{2}{4} = \frac{1}{2}$$

$$y = CF + PI$$

$$\boxed{y = \left(c_1 + c_2\frac{x^2}{2}\right)e^{-x^2} + \frac{1}{2}}$$

Q.11 Solve the following differential equation by the method of variation of parameter.

$$\frac{d^2y}{dx^2} + a^2y = \operatorname{cosec} ax$$

Sol^n ⇒ $\frac{d^2y}{dx^2} + a^2y = \operatorname{cosec} ax$ —— (1)

A.E. ⇒ $m^2 + a^2 = 0$

$m^2 = -a^2$

$\boxed{m = \pm ai}$

$$C.F. = c_1\cos ax + c_2\sin ax$$

Complete Solution ⇒

$$y = A \times \cos ax + B \times \sin ax \quad \text{—— (2)}$$

$u = \cos ax$, $v = \sin ax$

$\frac{du}{dx} = -a\sin ax$, $\frac{dv}{dx} = a\cos ax$

$$\cos ax\frac{dA}{dx} + \sin ax\frac{dB}{dx} = 0 \quad \text{—— (3)}$$

$$-a\sin ax\frac{dA}{dx} + a\cos ax\frac{dB}{dx} = \operatorname{cosec} ax \quad \text{—— (4)}$$

Eq^n (3) × $a\sin ax$ + Eq^n (4) × $a\cos ax$

$$a\sin ax\cos ax\frac{dA}{dx} + a\sin^2 ax\frac{dB}{dx} = 0$$

$$- \quad -a\sin ax\cos ax\frac{dA}{dx} + a\cos^2 ax\frac{dB}{dx} = \operatorname{cosec} ax \times a\cos ax$$

$$\cancel{a}\frac{dB}{dx} = \cancel{a}\cot ax$$

$$\frac{dB}{dx} = \cot ax$$

$$\int dB = \int \cot ax\, dx$$

$$\boxed{B = \frac{1}{a^2}\log|\sin ax| + c_1}$$

Put in the value of $\frac{dB}{dx}$ in Eq^n (3) ⟶

$$\cos ax\frac{dA}{dx} + \sin ax \times \cot ax = 0$$

$$\cos ax\frac{dA}{dx} = -\sin ax\cdot\cot ax$$

$$\frac{dA}{dx} = \frac{-\sin ax\cot ax}{\cos ax} = -\tan ax\cdot\cot ax$$

$$\frac{dA}{dx} = -1$$

$$\int dA = -\int 1\cdot dx$$

$$\boxed{A = -x + c_2}$$

Complete solution ⇒

$$\boxed{y = (-x + c_2) \times \cos ax + \frac{1}{a^2}\log|\sin ax| + c_1 \times \sin ax}$$

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Q.12 Solve the following differential equation by the method of variation of parameter.

$$\frac{d^2y}{dx^2} + a^2y = \sec ax$$

Sol^n ⇒ $\frac{d^2y}{dx^2} + a^2y = \sec ax$ — (1)

A.E. ⇒ $m^2 + a^2 = 0$

$m^2 = -a^2$

$\boxed{m = \pm ai}$

C.F. = $c_1 \cos ax + c_2 \sin ax$

C.S. ⇒ $y = A \times \cos ax + B \times \sin ax$ — (2)

$u = \cos ax$, $v = \sin ax$

$\frac{du}{dx} = -a \sin ax$, $\frac{dv}{dx} = a \cos ax$

$\cos ax \frac{dA}{dx} + \sin ax \frac{dB}{dx} = 0$ — (3)

$-a \sin ax \frac{dA}{dx} + a \cos ax \frac{dB}{dx} = \sec ax$ — (4)

Eq^n (3) $\times a \sin ax$ + Eq^n (4) $\times a \cos ax$

$\cancel{a \sin ax \cos ax \frac{dA}{dx}} + a \sin^2 ax \frac{dB}{dx} = 0$

$\cancel{-a \sin ax \cos ax \frac{dA}{dx}} + a \cos^2 ax \frac{dB}{dx} = \sec ax \times a \cos ax$

$a \frac{dB}{dx} = a$

$\cancel{a} \frac{dB}{dx} = \cancel{a}$

$\frac{dB}{dx} = 1$

$\int dB = \int 1\, dx$

$\boxed{B = x + c_1}$

Put in the value of $\frac{dB}{dx}$ in eq^n (3)

$\cos ax \frac{dA}{dx} + \sin ax = 0$

$\frac{dA}{dx} = -\tan ax$

$\int dA = -\int \tan ax \cdot dx$

$\boxed{A = \frac{-1}{a} \log |\sin ax| + c_2}$

Complete solution ⇒

$$\boxed{y = \left(\frac{-1}{a} \log|\sin ax| + c_2\right) \times \cos ax + (x + c_1) \times \sin ax}$$

Q.13 Solve the following differential equation

$$(x+1)^2 \frac{d^2y}{dx^2} - 3(x+1)\frac{dy}{dx} + 4y = x^2$$

Sol^n ⇒

$(x+1)^2 \frac{d^2y}{dx^2} - 3(x+1)\frac{dy}{dx} + 4y = x^2$ — (1)

Let $x + 1 = e^z$ ⇒ $z = \log(x+1)$

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$$(x+1)\frac{dy}{dx} = Dy$$

$$(x+1)^2\frac{d^2y}{dx^2} = D(D-1)y$$

$$D(D-1)y - 3(Dy) + 4y = x^2$$
$$D^2y - Dy - 3Dy + 4y = x^2$$
$$(D^2 - 4D + 4)y = x^2$$

Let $x+1 = e^z \Rightarrow x = e^z - 1$

$$(D^2 - 4D + 4)y = (e^z - 1)^2$$
$$(D^2 - 4D + 4)y = e^{2z} - 2e^z + 1$$

A.E. $\Rightarrow$ $m^2 - 4m + 4 = 0$

$(m-2)^2 = 0$

$\boxed{m = 2, 2}$

C.F. $= (c_1 + c_2 z)e^{2z}$

$$PI = \frac{1}{D^2-4D+4}(e^{2z}) - \frac{2}{D^2-4D+4}(e^z) + \frac{1}{D^2-4D+4}(1)$$

First term $= \frac{1}{(D-2)^2} e^{2z} = \frac{z^2}{2!}e^{2z} = \frac{z^2e^{2z}}{2}$

Second term, $D = 1$

$$\frac{-2}{1^2 - 4(1) + 4} e^z = \frac{-2}{1}(e^z) = -2e^z$$

For the third ~~therm~~ term, e^{0z} and substitude $D = 0$

$$\frac{1}{0^2 - 4(0) + 4} = \frac{1}{4}$$

So, the particular integral is:

$$y = \frac{z^2e^{2z}}{2} - 2e^z + \frac{1}{4}$$

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$y = CF + PI$

$$y = (c_1 + c_2 z)e^{2z} + \frac{z^2e^{2z}}{2} - 2e^z + \frac{1}{4}$$

$e^z = x+1$ and $z = \ln(x+1)$

$$\boxed{y = (c_1 + c_2\ln(x+1))(x+1)^2 + \frac{(\ln(x+1))^2(x+1)^2}{2} - 2(x+1) + \frac{1}{4}}$$

Q.14 Solve the following differential equations

$$x^2\left(\frac{d^2y}{dx^2}\right) - 5x\frac{dy}{dx} + 9y = x^3\log x$$

Sol^n $\Rightarrow$ $x^2\left(\frac{d^2y}{dx^2}\right) - 5x\frac{dy}{dx} + 9y = x^2\log x$ —— (1)

$$(D'(D'-1) - 5(D') + 9)y = x^2\log x$$
$$(D'^2 - D' - 5D' + 9)y = x^2\log x$$
$$(D'^2 - 6D' + 9)y = x^2\log x$$

A.E. $\Rightarrow$ $m^2 - 6m + 9 = 0$

$m^2 - 3m - 3m + 9 = 0$

$(m-3)(m-3) = 0$

$\boxed{m = 3, 3}$

C.F. $= (c_1 + c_2 z)e^{3z}$

Let $x = e^z$, $\Rightarrow z = \log x$ and $x^2 = e^{2z}$

$$P.I. = \frac{1}{D^2 - 6D + 9}(e^{2z})z = \frac{1}{(D-3)^2}(e^{2z})z$$

$$PI = e^{2z}\frac{1}{(D+2-3)^2}z = e^{2z}\frac{1}{(D-1)^2}z$$

$$PI = e^{2z}(D-1)^{-2}z = e^{2z}(1 + 2D + 3D^2 + \dots)z$$

$$PI = e^{2z}(z + 2Dz + 0)$$

$$PI = e^{2z}(z + 2)$$

$$PI = x^2(\log x + 2)$$

$y = CF + PI$

$$\boxed{y = (c_1 + c_2 \log x)x^3 + x^2(\log x + 2)}$$

Q. 15 Solve the following differential equation by Charpit's Method : $p = (qy+z)^2$

Sol^n ⇒ $p = (qy+z)^2$

$p = q^2y^2 + 2qyz + z^2$

$p - q^2y^2 - 2qyz - z^2 = 0$

$\frac{\delta f}{\delta x} = 0$, $\frac{\delta f}{\delta y} = -2q^2y$ $\frac{\delta f}{\delta z} = -2qy - 2z$ $\frac{\delta f}{\delta p} = 1$

$\frac{\delta f}{\delta q} = -2qy^2 - 2yz$

$$\frac{dp}{-p(2qy+2z)} = \frac{dq}{-2q^2y - q(2qy+2z)} = \frac{dz}{-p(1)+q(2qy^2+2yz)}$$

$$= \frac{dx}{-1} = \frac{dy}{2qy^2 + 2yz}$$

$$\frac{dp}{-2p(qy+z)} = \frac{dq}{-4q(qy+z)}$$

$$\frac{dp}{-2p} = \frac{dq}{-4q}$$

$$\frac{dp}{+p} = \frac{dq}{2q}$$

$$\log p = \frac{1}{2}\log q$$

$$\boxed{p = \sqrt{q}}$$

?

$$\frac{dp}{-2p(qy+z)} = \frac{dy}{2y(qy+z)}$$

$$\frac{dp}{-2p} = \frac{dy}{2y}$$

$\log p + \log y = \log c$

$py = c$

$$\boxed{p = c/y}$$

$p = (qy+z)^2$

$\frac{c}{y} = (qy+z)^2$

$\sqrt{\frac{c}{y}} = qy + z$

$q = \frac{\sqrt{c} - z\sqrt{y}}{y^{3/2}}$

$dz = p\,dx + q\,dy$

$dz = \frac{c}{y}dx + \frac{\sqrt{c}}{y^{3/2}}dy - \frac{z}{y}dy$

$dz + \frac{z}{y}dy = \frac{a}{y}dx + \frac{\sqrt{a}}{y^{3/2}}dy$

$y\,dz + z\,dy = a\,dx + \sqrt{a}\,y^{-1/2}dy$

$d(yz) = a\,dx + \sqrt{a}\,y^{-1/2}dy$

$\int d(yz) = \int a\,dx + \int \sqrt{a}\cdot y^{-1/2}dy$

$yz = ax + \sqrt{a}\,\frac{y^{1/2}}{1/2} + b$

$$\boxed{yz = ax + 2\sqrt{ay} + b}$$

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Q.16 Solve the following differential equation:

$$z - xp - yq = a\sqrt{x^2+y^2+z^2}$$

Sol^n ⇒

$$z - xp - yq = a\sqrt{x^2+y^2+z^2}$$

$$z = xp + yq + a\sqrt{x^2+y^2+z^2} \quad \text{—— (1)}$$

$$z - a\sqrt{x^2+y^2+z^2} = xp + yq \quad \text{—— (2)}$$

$$z - a(x^2+y^2+z^2)^{1/2} = xp + yq$$

$$\underset{(3)}{\frac{dx}{x}} = \underset{(4)}{\frac{dy}{y}} = \underset{(5)}{\frac{dz}{z - a(x^2+y^2+z^2)}},$$

Taking multiplier x, y, z ⇒ $x^2+y^2+z^2 = t^2$

$$\frac{x\,dx + y\,dy + z\,dz}{x^2+y^2+z^2 - az(x^2+y^2+z^2)^{1/2}}$$

$$\frac{t\,dt}{t^2 - azt} = \frac{at}{t - az} \quad \text{—— (6)}$$

Taking multiplier $0, 0, 1, 1$

$$\frac{dz + dt}{z - at + t - az} = \frac{dz + at}{z + t - a(z+t)}$$

$$\frac{1}{y - a} = \frac{az + at}{z + t} \quad \text{—— (7)}$$

Eq^n (3) and (4)

$$\frac{dx}{x} = \frac{dy}{y}$$

$$\log x = \log y + c_1$$

$$\frac{x}{y} = c_1$$

$$\boxed{u = x/y}$$

Eq^n (3) and (7) —

$$\frac{dx}{x} = \frac{az + at}{(1-a)(z+t)}$$

$$\log x = \frac{1}{(1-a)}\log|z+t| + \log c_2$$

$$\frac{x(1-a)}{z+t} = c_2 \Rightarrow \boxed{V = \frac{x+a}{z+t}}$$

$$\boxed{f\left(\frac{x}{y}, \frac{x(1-a)}{z+\sqrt{x^2+y^2+z^2}}\right) = 0}$$

Q.17 Solve the following differential equations:

$$\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}$$

$$\Rightarrow \underset{1}{\frac{dx}{mz - ny}} = \underset{2}{\frac{dy}{nx - lz}} = \underset{3}{\frac{dz}{ly - mx}}$$

$$x(mz - ny) + y(nx - lz) + z(ly - mx) = 0$$

∵ Taking multiplier x, y, z

$$\frac{dy}{nx - lz} = x\,dx + y\,dy + z\,dz$$

$$\int x\cdot dx + \int y \cdot dy + \int z \cdot dz = 0$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = 0$$

$$l(mz - ny) + m(nx - lz) + n(ly - mx) = 0$$

$$\frac{dx}{mz - ny} = l\,dx + m\,dy + n\,dz$$

$$\int l \cdot dx + \int m \cdot dy + \int n \cdot dz = 0$$

$$lx + my + nz = 0$$

$$f(u, v) = 0$$

$$\boxed{f(x^2+y^2+z^2,\ lx + my + nz) = 0}$$

Q.18 Solve the following differential equation by Charpit's method

$$q = px + p^2 \quad \text{— (1)} \qquad dz = p\,dx + q\,dy \quad \text{— (2)}$$

Diff. (1) w. r. to x, y, z, p and q (partially)

$$\frac{\delta f}{\delta x} = p, \quad \frac{\delta f}{\delta y} = 0, \quad \frac{\delta f}{\delta z} = 0, \quad \frac{\delta f}{\delta p} = x + 2p$$

$$\frac{\delta f}{\delta q} = -1$$

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$$\frac{dp}{\frac{\delta f}{\delta x}+p\frac{\delta f}{\delta z}} = \frac{dq}{\frac{\delta f}{\delta y}+q\frac{\delta f}{\delta z}} = \frac{dz}{-p\frac{\delta f}{\delta p}-q\frac{\delta f}{\delta q}} = \frac{dx}{-\frac{\delta f}{\delta p}} = \frac{dy}{-\frac{\delta f}{\delta q}} = \frac{df}{c}$$

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$$\frac{dx}{-(x+2p)} = \frac{dy}{-(-1)} = \frac{dz}{-p(x+2p)-q(-1)} = \frac{dp}{p+p(0)} = \frac{dq}{0+q(0)}$$

$$\frac{dx}{-(x+2p)} = \frac{dy}{1} = \frac{dz}{-px-2p^2+q} = \frac{dp}{p} = \frac{dq}{0}$$

$dq = 0$ means that q is a constant

$q = a$, where a is an arbitrary constant

$\boxed{q=a}$

$q = px + p^2$

$a = px + p^2$

$p^2 + px - a = 0$

$p = \frac{-x \pm \sqrt{x^2+4a}}{2}$ → Put in Eq^n (2)

$dz = p\,dx + q\,dy$

$dz = \left(\frac{-x \pm \sqrt{x^2+4a}}{2}\right)dx + a\,dy$

$dz = \left(\frac{-x}{2} \pm \frac{\sqrt{x^2+4a}}{2}\right)dx + a\,dy$

$dz = \frac{-x}{2}dx \pm \frac{1}{2}\sqrt{x^2+4a}\,dx + a\,dy$

$\int dz = \int \frac{-x}{2}dx \pm \int \frac{1}{2}\sqrt{x^2+4a}\,dx + \int a\cdot dy$

$$\boxed{z = \frac{-x^2}{4} \pm \frac{x}{4}\sqrt{x^2+4a} + a\ln|x| + \sqrt{x^2+4a} + C + ay}$$

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Q.19 Solve the following partial differential equation:

$p\tan x + q\tan y = \tan z$

⇒ $p\tan x + q\tan y = \tan z$

$$\frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}$$

(1) (2) (3)

Taking (1) and (2) —

$\frac{dx}{\tan x} = \frac{dy}{\tan y}$

$\int \sec(x)dx = \int \sec y\,dy$

$\log|\tan x| = \log|\tan y| + C_1$

$\log|\tan x| - \log|\tan y| = C_1$

$$\boxed{C_1 = \frac{\tan x}{\tan y}}$$

Taking (2) and (3) —

$\frac{dy}{\tan y} = \frac{dz}{\tan z}$

$\sec y\,dy = \sec z\,dz$

$\log|\tan y| = \log|\tan z| + C_2$

$\log|\tan y| - \log|\tan z| = C_2$

$$\boxed{\frac{\tan y}{\tan z} = C_2}$$

$$\boxed{f\left[\frac{\tan x}{\tan y}, \frac{\tan y}{\tan z}\right] = 0}$$

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Q.20 Solve the following partial differential equation
$$xzp + yzq = xy$$

Sol^n ⇒ $xzp + yzq = xy$

Lagrange's auxiliary equation —
$$\frac{dx}{xz} = \frac{dy}{yz} = \frac{dz}{xy}$$
1. 2. 3.

Taking eq^n (1) and (2) —
$$\frac{dx}{x\cancel{z}} = \frac{dy}{y\cancel{z}}$$
$$\frac{dx}{x} = \frac{dy}{y}$$
$$\log|x| = \log|y| + C_1$$
$$\log|x| - \log|y| = C_1$$
$$\boxed{\log\left(\frac{x}{y}\right) = C_1}$$

Taking (1) and (3) —
$$\frac{dx}{\cancel{x}z} = \frac{dz}{\cancel{x}y}$$
$$y\,dx = dz \cdot z$$
$$\int y\,dx = \int z\,dz$$
$$yx = \frac{z^2}{2} + C_2$$
$$\boxed{2yx - z^2 = C_2}$$
$$\boxed{f\left[\log\left(\frac{x}{y}\right), 2yx - z^2\right]}$$

Q.21 Show that there are 8 points of intersection of the biquadratic curve
$(x^2-4y^2)(x^2-9y^2) + 5x^2y - 5xy^2 - 30y^3 + xy + 7y^2 - 1 = 0$
and it asymptotes which lie on a circle.

Sol^n ⇒ given equation ⇒
$$(x^2-4y^2)(x^2-9y^2) + 5x^2y - 5xy^2 - 30y^3 + xy + 7y^2 - 1 = 0$$
$$\phi_4(m) = (1-4m^2)(1-9m^2)$$
$$= 1 - 13m^2 + 36m^4$$
$$\phi'_4(m) = -26m + 144m^3$$
$$\phi_3(m) = 5m - 5m^2 - 30m^3$$
$$\phi_2(m) = -m + 7m^2$$
$$\because \phi_4(m) = 0$$
$$(1-4m^2)(1-9m^2) = 0$$
$$(1+2m)(1-2m)(1+3m)(1-3m) = 0$$
$$m = -\frac{1}{2}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{2}$$
$$c = \frac{-\phi_3(m)}{\phi'_4(m)} \Rightarrow c = \frac{-(5m - 5m^2 - 30m^3)}{-26m + 144m^3}$$
$$c = \frac{-5m(1-m-6m^2)}{2m(72m^2-13)}$$
$$c = \frac{-5}{2}\frac{(1-3m)(1+2m)}{(72m^2-13)}$$

When $m = -\frac{1}{3} \Rightarrow c = \frac{-5}{2}\frac{(2)(1/3)}{(72\times 1/9 - 13)} = \frac{1}{3}$

$m = -\frac{1}{3}$, $c = +\frac{1}{3}$

When $m = -\frac{1}{2}$, $c = 0$

When $m = \frac{1}{3}$, $c = 0$ and when $m = \frac{1}{2}$, $\boxed{c = \frac{1}{2}}$

Eq^n of the line is →

$y = mx + c$

$\boxed{3y + x - 1 = 0}$ — (1) , $\boxed{2y + x = 0}$ — (2)

$\boxed{3y - x = 0}$ — (3) and $\boxed{2y - x - 1 = 0}$ — (4)

→ Multiply the equation (1), (2), (3) and (4)

$A = (3y+x-1)(2y+x)(3y-x)(2y-x-1) = 0$

$A = (3y-x)(3y+x-1)(2y+x)(2y-x-1) = 0$

$A = (9y^2 - x^2 - 3y + x)(4y^2 - x^2 - 2y - x) = 0$

$A = (36y^4 - 13x^2y^2 + x^4) + (-30y^3 - 5xy^2 + 5x^2y)$
$+ (6y^2 + xy - x^2) = 0$ — (B)

Taking Eq^n (A) and (C)

$C = (x^2 - 4y^2)(x^2 - 9y^2) + 5x^2y - 5xy^2 - 30y^3 + xy + 7y^2 - 1 = 0$

$C = x^4 - 9x^2y^2 - 4x^2y^2 + 36y^4 + 5x^2y - 5xy^2 - 30y^3 + xy + 7y^2 - 1 = 0$

$C - A = 0$

$y^2 + x^2 - 1 = 0 \Rightarrow \boxed{x^2 + y^2 = 1}$

No. of intersection are $n(n-2)$

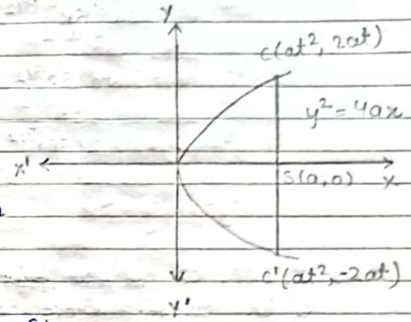
$4(4-2) = 8$

Radius of the circle is intersection lies on the eight point.

$\boxed{x^2 + y^2 = 1}$

H.P.

Q. 22 Show that in the parabola $y^2 = 4ax$, the radius of curvature at any point P is $\frac{2(SP)^{1/2}}{\sqrt{a}}$, where S is the focus of the parabola.



Sol^n →

$y^2 = 4ax$ — (1)

$x = at^2$, $y = 2at$

$$\rho = \frac{[\dot{x}^2 + \dot{y}^2]^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$\frac{dx}{dt} = 2at$, $\frac{d^2x}{dt^2} = 2a$

$\frac{dy}{dt} = 2a$; $\frac{d^2y}{dt^2} = 0$

$$\rho = \frac{[4a^2t^2 + 4a^2]^{3/2}}{2at(0) - 2a(2a)}$$

$$= \frac{[4a^2t^2 + 4a^2]^{3/2}}{-4a^2}$$

$$\rho = \frac{((2)^2a^2)^{3/2}(t^2+1)^{3/2}}{-4a^2}$$

$$\rho = \frac{8a^3(1+t^2)^{3/2}}{-4a^2}$$

$\rho = 2a(1+t^2)^{3/2}$ — (2)

$SP = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$SP = \sqrt{(at^2 - a)^2 + (2at - 0)^2}$

$SP = \sqrt{a^2(t^2-1)^2 + 4a^2t^2}$

$SP = a\sqrt{t^4 - 2t^2 + 1 + 4t^2}$

$SP = a\sqrt{t^4 + 2t^2 + 1} = a(t^2+1)$

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$\frac{SP}{a} = (t^2+1)$ — (3)

from eq^n (2) and (3) →

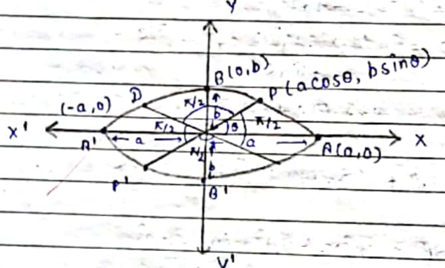
$\rho = 2a\left[\frac{SP}{a}\right]^{3/2}$

$\rho = \frac{2a\, SP^{3/2}}{a^{3/2}}$

$$\boxed{\rho = \frac{2(SP)^{3/2}}{\sqrt{a}}}$$

H.P.

Q.23 If CP, CD be a pair of conjugate semi-diameter of an ellipse. Prove that the radius of curvature ρ is $\frac{(CD)^3}{ab}$, where a and b are the length of the semi-axes of the ellipse.



Equation of the ellipse is →

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ — (1)

$x = a\cos\theta$ — (2) $\qquad y = b\sin\theta$ — (3)

$\dot{x} = -a\sin\theta \qquad \dot{y} = b\cos\theta$

$\ddot{x} = -a\cos\theta \qquad \ddot{y} = -b\sin\theta$

$\rho = \frac{[\dot{x}^2 + \dot{y}^2]^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$

$= \frac{[a^2\sin^2\theta + b^2\cos^2\theta]^{3/2}}{a\sin\theta \times b\sin\theta + b\cos\theta \times a\sin\theta}$

$= \frac{[a^2\sin^2\theta + b^2\cos^2\theta]^{3/2}}{ab\sin^2\theta + ba\cos^2\theta}$

$\rho = \frac{[a^2\sin^2\theta + b^2\cos^2\theta]^{3/2}}{ab}$ — (A)

$P(x_1, y_1) \quad Q(x_2, y_2)$

P ———— Q

by distance formula

$PQ = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$CD = \sqrt{(-a\sin\theta - 0)^2 + (b\cos\theta - 0)^2}$

$CD = \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}$

$(CD)^2 = a^2\sin^2\theta + b^2\cos^2\theta$ — (B)

from equation (A) and (B) →

$\rho = \frac{[CD^2]^{3/2}}{ab}$

$$\boxed{\rho = \frac{[CD]^3}{ab}}$$

H.P.

Q.24 Solve the following differential equation

$(x^4y^4 + 5xy + 2)y\,dx + (x^2y^2 + 4xy + 2)x\,dy = 0$

Sol^n → $(x^4y^5 + 5xy^2 + 2y)dx + (x^3y^2 + 4x^2y + 2x)dy = 0$

$M = x^4y^5 + 5xy^2 + 2y$

$N = x^3y^2 + 4x^2y + 2x$

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$$\frac{\delta M}{\delta y} = 5x^4y^4 + 10xy + 2$$

$$\frac{\delta N}{\delta x} = 3x^2y^2 + 8xy + 2$$

$$\frac{1}{N}\left(\frac{\delta M}{\delta y} - \frac{\delta N}{\delta x}\right)$$

$$\frac{1}{x^3y^2+4x^2y+2x}\left(5x^4y^4 + 10xy + 2 - 3x^2y^2 + 8xy - 2\right)$$

$$\frac{1}{x^3y^2+4x^2y+2x}\left(5x^4y^4 + 2xy - 3x^2y^2\right)$$

$$\frac{5x^4y^4 + 2xy - 3x^2y^2}{x^3y^2 + 4x^2y + 2x}$$

$$\text{I.F.} = \frac{1}{Mx - Ny}$$

$$\text{IF} = \frac{1}{x^5y^5 + 5x^2y^2 + 2xy - (x^3y^3 + 4x^2y^2 + 2xy)}$$

$$\text{I.F.} = x^5y^5 - x^3y^3 + x^2y^2$$

I.F x (1)

$$x^3y^2[xy^3\,dx + dy) + xy(5y\,dx + 4x\,dy) + 2(y\,dx + x\,dy) = 0$$

$$\int d\frac{(x^5y^5)}{5} + \int d(x^5y^4) + \int d(x^2y^2) = 0$$

$$\boxed{\frac{x^5y^5}{5} + x^5y^4 + x^2y^2 = 0}$$

Q.25 Solve the following differential equation

$$(xy + 2x^2y^2)\,y\,dx + (xy - x^2y^2)\,x\,dy = 0$$

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Sol^n ⇒ $(xy^2 + 2x^2y^3)\,dx + (x^2y - x^3y^2)\,dy = 0$

$$M = xy^2 + 2x^2y^3$$

$$N = x^2y - x^3y^2$$

$$\frac{\delta M}{\delta y} = 2xy + 6x^2y^2$$

$$\frac{\delta N}{\delta x} = 2xy - 3x^2y^2$$

$$\text{I.F} = \frac{1}{Mx - Ny}$$

$$\text{IF} = \frac{1}{3x^3y^3}$$

$$\frac{1}{3x^3y^3}(xy^2 + 2x^2y^3)\,dx + \frac{1}{3x^3y^3}(x^2y - x^3y^2)\,dy$$

$$\left(\frac{1}{3x^2y} + \frac{2}{3x}\right)dx + \left(\frac{1}{3xy^2} - \frac{1}{3y}\right)dy = 0$$

$$\frac{\delta M'}{\delta y} = \frac{-1}{3x^2y^2}, \quad \frac{\delta N'}{\delta x} = \frac{-1}{3x^2y^2}$$

$$\int \frac{1}{3x^2y} + \frac{2}{3x}\,dx + \int \frac{-1}{3y}\,dy$$

$$\frac{-1}{3xy} + \int \frac{2}{3x}\,dx + \int \frac{-1}{3y}\,dy$$

$$\frac{-1}{3xy} + \frac{2}{3}\log x - \frac{1}{3}\log y$$

$$\boxed{\frac{-1}{xy} + 2\log x - \log y}$$

B

Q.26 Find the point where the value of $u = x^3 + y^3 - 3axy$ is minimum or maximum.

Sol^n ⇒

$$\frac{\delta f}{\delta x} = 3x^2 - 3ay, \quad \frac{\delta f}{\delta y} = 3y^2 - 3ax$$

$$A = \frac{\delta^2 f}{\delta x^2} = 6x, \quad C = \frac{\delta^2 f}{\delta y^2} = 6y$$

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$B = \frac{\delta^2 f}{\delta x \delta y} = -3a$

for max. or mini.

$\frac{\delta f}{\delta x} = 0 \quad , \quad \frac{\delta f}{\delta y} = 0$

$3x^2 - 3ay = 0$

$x^2 - ay = 0$

$x^2 = ay$

$x^2 - ay = 0$ — (1)

$3y^2 - 3ax = 0$

$y^2 = ax$

$y^2 - ax = 0$ — (2)

$\boxed{y^2 = ax}$

$\left(\frac{x^2}{a}\right)^2 = ax$

$x^4 = a^3 x$

$x(x^3 - a^3) = 0$

$x = 0 \quad (x^3 - a^3) = 0$

$\boxed{x = 0} \quad (x-a)\underbrace{(x^2 + ax + a^2)}_{\text{Imaginary}}$

$x = 0, \; x = a$. Put in equation (2)

$y = 0$

$y^2 = a^2 \Rightarrow \boxed{y = \pm a}$

$x = 0, y = 0$

$O(0,0) \; , \; P(a,a) \; , \; Q(a,-a)$

Consider $\rightarrow AC - B^2 < 0$

$0 - 9a^2 = -9a^2$

$AC - B^2 < 0$

Neither max. nor min. at $(0,0)$

(2) $A = 6a, \quad C = 6a$

$AC - B^2$

$36a^2 - 9a^2 = 27a^2$

$AC - B^2 > 0$ (minima)

3. $AC - B^2 \Rightarrow -36a^2 - 9a^2 = -45a^2$

Neither maxima nor minima

Q.27 Show that $u = x^2y^2(1-x-y)$ is maximum at $x = 1/3, y = 1/3$.

$u = x^2y^2(1-x-y)$

$u = x^2y^2 - x^3y^2 - x^2y^3$

$\frac{\delta u}{\delta x} = 2xy^2 - 3x^2y^2 - 2xy^3$

$= xy^2(2 - 3x - 2y)$

$\frac{\delta u}{\delta y} = 2x^2y - 2x^3y - 3x^2y^2$

$x^2y(2 - 2x - 3y)$

$xy^2(2 - 3x - 2y) = 0$

$x^2y(2 - 2x - 3y) = 0$

$2 - 3x - 2y = 0 \times 3$

$2 - 2x - 3y = 0 \times 2$

$6 - 9x - \cancel{6y} = 0$

$\underline{\overset{-}{4} \overset{+}{-} 4x \overset{+}{-} \cancel{6y} = 0}$

$2 - 5x = 0$

$\boxed{x = 2/5} \; , \; \boxed{y = 2/5}$

Let assume the function

$u = x^2y^2(6 - 3x - 2y)$

$\frac{\delta u}{\delta x} = y^2(6 - 3x - 2y)(2) + x^2y^2(-3)$

$\frac{\delta u}{\delta y} = x^2(6 - 3x - 2y)(1) + x^2y^2(-2)$

$12 - 9x - \cancel{4y} = 0$

$\underline{6 \overset{+}{-} 3x \overset{+}{-} \cancel{4y} = 0}$

$6 - 6x = 0$

$\boxed{x = 1} , \; \boxed{y = 3/4}$

Q.28 Trace the curve $y^2(a+x) = x^2(a-x)$

$\Rightarrow$ $f(x,y) = y^2(a+x) - x^2(a-x)$

$f(x,-y) = (-y)^2(a+x) - x^2(a-x)$

$f(x,y) = f(x,-y)$

1. Symmetric about x-axis
2. At origin $0 = 0$ satisfied
3. Tangent at origin $(0,0)$

$ay^2 + xy^2 - ax^2 + x^3$

$a(y^2 - x^2) = 0$

$\boxed{y = \pm x}$

4. Intersection at axes —

(i) x-axis

$y = 0$ $\quad O(0,0)$

$0 = -x^2(a-x)$ $\quad A(a,0)$

$x = 0,\ x = a$

(ii) y-axis

$x = 0$

$y^2 a = 0$ $\quad O(0,0)$

$y = 0$

5. Asymptotes

(i) $\parallel$ to x-axis

$-1 = 0$ (N.P.)

(ii) $\parallel$ to y-axis

$(a+x) = 0$ (N.P.)

(iii) Oblique $\rightarrow$ Put $x = 1$; $y = m$

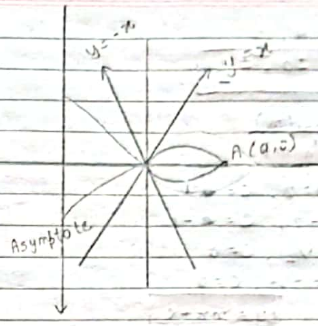
$\phi_3(m) = m^2 + 1 \Rightarrow \phi_3'(m) = 2m$

$\phi_2(m) = m^2 a - a \Rightarrow a(m^2 - 1)$

$\because \phi_3(m) = 0$

$m^2 + 1 = 0$

$m = \pm\sqrt{i}$ (N.P.) $\rightarrow$ imaginary



Q.29 Find the asymptotes of the curve $4(x^4 + y^4) - 17x^2y^2 - 4x(4y^2 - x^2) + 2(x^2 - 2) = 0$ and show that they pass through the point of intersection of the curve with the ellipse $x^2 + 4y^2 = 4$

Sol^n $\Rightarrow$

$4(x^4 + y^4) - 17x^2y^2 - 4x(4y^2 - x^2) + 2(x^2 - 2) = 0$

$4x^4 + 4y^4 - 17x^2y^2 - 16xy^2 + 4x^3 + 2x^2 - 4 = 0$

$\phi_4(m) = 4 + 4m^4 - 17m^2$

$\phi_3(m) = -16m^2 + 16$

$\phi_3(m) = 16(m^2 - 1)$

$\phi_2(m) = 0$

$\phi_4'(m) = 16m^3 - 34m$

$\because \phi_4(m) = 0$

$4m^4 - 17m^2 + 4 = 0$

let $m^2 = t$

$4t^2 - 17t + 4 = 0$

$t = \dfrac{17 \pm \sqrt{289 - 4(16)}}{8}$

$t = \dfrac{17 \pm \sqrt{225}}{8} \Rightarrow t = \dfrac{17 \pm 15}{8}$

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$t = 4, \frac{1}{4}$

$\boxed{m = \pm 2, \pm\frac{1}{2}}$

$C = \frac{-\Phi_{n-1}(m)}{\Phi'_n(m)}$

When $m = 2,\ C = 1$

$m = -2,\ C = -1$

$m = 1/2,\ C = 0$

$m = -1/2,\ C = 0$

Eq^n as $\rightarrow$ $\boxed{y = mx + c}$

$2y = x$ — (1) $\qquad 2y = -x$ — (2)

$y = 2x+1$ — (3) $\qquad y = -2x-1$ — (4)

from Eq^n (1), (2), (3), (4) $\rightarrow$

$(2x-y+1)(2x+y+1)(x-2y)(x+2y)$ — (B)

$+2x+y+1+2x(2x+y+1)-y(2x+y+1) \quad x(x+2y)-2y(x+2y)$

$2x+y+1+2x-y \quad 4x^2+2xy-2xy-y^2+x^2+2yx-2yx-4y^2$

$4x \quad 4x^2-4y^2-y^2+x^2 = 0$ — (B)

Taking Eq^n (A) & (B) $\rightarrow$

$\boxed{x^2 + 4y^2 = 4}$ H.P.

Q.30 Trace the curve $x^3 + y^3 = 3xy$

$\Rightarrow$ $f(x,y) = x^3 + y^3 - 3xy$

$f(y,x) = y^3 + x^3 - 3xy$

$f(x,y) = f(y,x)$

1. Symmetric about $y = x$.
2. Origin $0 = 0$ satisfied
3. Tangent at origin

$x^3 + y^3 - 3xy = 0$

$x = 0,\ y = 0$

Intersection at axis

$\parallel$ to x-axis $\qquad \parallel$ to y-axis $x = 0$

$y = 0 \Rightarrow x_1 = 0 \qquad y = 0$

(V) Oblique $x = 1,\ y = m$

$\Phi_3(m) = m^3 + 1$

$\Phi_2(m) = -3m$

$\because \Phi_3(m) = 0$

$m^3 + 1 = 0$

$\boxed{m = -1}$

Put in (I) $y = x$

$x^3 + x^3 = 3x^2$

$2x^3 = 3x^2$

$x = \frac{3}{2}$

$P\left(\frac{3}{2}, \frac{3}{2}\right)$

$m = \frac{1 \pm \sqrt{3}i}{2}$

$c = \frac{-\Phi_{n-1}(m)}{\Phi_n'(m)} \Rightarrow C = \frac{-\Phi_2(m)}{\Phi_3'(m)}$

$C = \frac{-3m}{3m^2}$

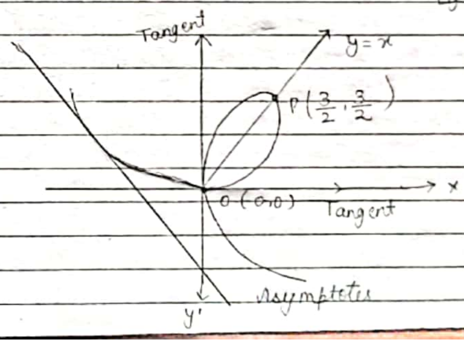
When $m = -1 \Rightarrow C = \frac{-3}{3} = -1$

$\boxed{C = -1}$

$\boxed{y = mx + c}$

$y = -x + (-1) \Rightarrow \boxed{y = -x - 1}$

$\boxed{y + x + 1}$



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